

Titration Curve Lab

Where the equivalence point actually sits, why the volume and the pH answer to two different questions, and what the indicator has to match.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/titration-curve/

1 Before you open anything

Answer from what you already know. Nothing here is graded on being right — the point is to have a committed answer on paper before the curve settles it.

PREDICTION

You titrate 25.0 mL of 0.100 M acetic acid (CH_3COOH , $\text{pK}_a = 4.75$) with 0.100 M NaOH. At the equivalence point — the moment every mole of acid has been neutralised — what is the pH? Give a number, not a range.

Now the same question for 25.0 mL of 0.100 M HCl titrated with 0.100 M NaOH. pH at equivalence?

Write one or two sentences on your reasoning — specifically, what you assumed is left in the flask at the equivalence point.

2 Guided setup: the strong-acid run

Open the demonstration. Build the strong-acid curve first, so you have a reference shape to compare everything else against.

1. In the **In the flask (analyte)** panel, click **Strong acid**. The hint underneath should now name HCl.
2. Set **Analyte concentration** to **0.100 M** and **Analyte volume** to **25.0 mL**.
3. In **In the burette (titrant)**, confirm **NaOH (strong base)** is selected and set **Titrant concentration** to **0.100 M**.

4. Before adding anything, read and record the **Equivalence vol.** and **pH at equivalence** boxes.
5. Now add titrant with the **+5.00 mL**, **+1.00 mL** and **+0.10 mL** buttons to hit each volume in the table, reading **pH now** at each one. Check the **Titrant added** box to be sure you are where you think you are; **Undo** steps back if you overshoot.

a. Equivalence vol. = _____ mL pH at equivalence = _____

Titrant added (mL)	pH now	Change in pH since the row above
0.00		—
10.00		
20.00		
24.00		
24.90		
25.10		
30.00		
50.00		

b. Between which two rows is the pH change largest? State the pH change and the volume of NaOH that produced it.

c. The hint under the delivery buttons notes that one drop is roughly 0.05 mL. Using your answer to **b**, argue whether a single drop near equivalence matters more or less than a single drop at 10.00 mL, and by roughly how much.

Moving any slider empties the flask and restarts the run. Finish a curve before you change anything in the setup panels.

3 The core relationship: acid strength and the equivalence pH

Now switch the flask to a weak acid and change one thing only — how weak it is. Everything else stays fixed, so anything that moves is caused by pK_a alone.

1. Click **Weak acid**. Keep **Analyte concentration** at **0.100 M**, **Analyte volume** at **25.0 mL**, and the burette at **NaOH, 0.100 M**.
2. Set the **Acid strength, pK_a** slider to each value in the table. The hint underneath names a real acid when you land on one of these.
3. For each pK_a , read the **Equivalence vol.** box and the **Key points on this curve** table at the bottom of the controls. You do not need to add any titrant to fill this in.

pK_a	Acid named in the hint	Equivalence vol. (mL)	pH at Start	pH at Half-equivalence	pH at Equivalence
3.17					
4.75					
6.35					
7.53					
9.25					

- a. One column is identical in every row. Which one? Explain why changing pK_a cannot change it.

- b. Compare the half-equivalence column with the pK_a column. State the relationship. (The two agree to within about 0.02 for most rows and drift slightly apart for the strongest of these weak acids — the demonstration solves the full charge balance rather than using the buffer approximation.)

c. Write the rule connecting pK_a to the pH at equivalence, in your own words, and give the chemical reason. Name the species responsible.

d. Go back to your two predictions in Part 1. Which one was right, which was not, and what exactly did you assume that turned out to be incomplete?

4 Using it the way a question would

4A — Working backwards to a concentration. A titration is run to find an unknown, so the volume is the measurement and the concentration is the answer.

1. Set the flask to **Weak acid, Analyte volume 20.0 mL, Analyte concentration 0.150 M, pK_a 5.50**, burette **NaOH 0.100 M**.
2. *Before* looking at the readout, calculate the equivalence volume on paper. Write it below, then check it against **Equivalence vol.**

a. Your calculated equivalence volume: _____ mL Readout: _____ mL.
Show the calculation.

b. From the **Key points on this curve** table, record the half-equivalence volume and pH. If this were an unknown acid and those were your only two measurements — equivalence volume and half-equivalence pH — show how you would recover both its concentration and its K_a .

4B — Where a habit breaks. A common shortcut is to treat indicator choice as a matter of preference, since any of them changes colour somewhere on an acid–base titration. Test that.

1. Return to **Weak acid, pK_a 4.75, 0.100 M, 25.0 mL**, burette **NaOH 0.100 M**.
2. Work through the **Indicator** menu and record **Colour changes at** and **Titration error** for each one.
3. Then click **Strong acid** (same concentrations and volume) and repeat.

Indicator (pH range)	Weak acid: colour changes at (mL)	Weak acid: titration error	Strong acid: colour changes at (mL)	Strong acid: titration error
Methyl orange (3.1–4.4)				
Methyl red (4.4–6.2)				
Bromothymol blue (6.0–7.6)				
Phenolphthalein (8.3–10.0)				

c. Methyl orange is nearly harmless on one of these titrations and catastrophic on the other. Give both errors, and explain the difference by referring to where the indicator's pH range falls on each curve.

d. If you used methyl orange on the weak acid and reported the acid's concentration from the volume at which the colour changed, would your reported value be too high or too low? By what percentage? Justify the direction, not just the number.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, set up each scenario, and check yourself – if a number is off, write down where the method went wrong. Assume $25\text{ }^{\circ}\text{C}$, $K_{\text{W}} = 1.00 \times 10^{-14}$, throughout.

1. 25.0 mL of 0.100 M HCl is titrated with 0.100 M NaOH. Calculate the pH after exactly 24.00 mL of NaOH has been added. Report to two decimal places.

2. 25.0 mL of 0.100 M of a monoprotic weak acid HA ($K_{\text{a}} = 1.8 \times 10^{-5}$) is titrated with 0.100 M NaOH. Calculate the pH (a) after 10.00 mL of NaOH, and (b) at the equivalence point. State which equilibrium controls each answer.

3. A 20.0 mL sample of a monoprotic weak acid of unknown concentration requires 30.00 mL of 0.100 M NaOH to reach the equivalence point. A pH meter reads 5.50 when 15.00 mL of NaOH has been delivered. Find (a) the concentration of the acid, (b) its K_{a} , and (c) the pH at the equivalence point.

4. Two flasks each hold 25.0 mL of a 0.100 M monoprotic acid — one strong, one weak. Both are titrated with 0.100 M NaOH. Explain, without calculating, why the two titrations reach equivalence at the same volume but not at the same pH. Then state which of the two starting pH values is higher, and why.

6 On your own

Nothing here is graded, and the demonstration does considerably more than this worksheet uses.

- Tick **Reveal the full calculated curve** and press **Run slowly**. Watch where the recorded points bunch up and where they spread out, and work out what that spacing is telling you about the rate of pH change.
 - Put **Strong base** in the flask — the burette switches to HCl. Sketch the curve you expect before you add anything, then build it. Where is equivalence, and why?
 - Set the analyte to a strong acid and then click **HCl (strong acid)** in the burette as well. Read the warning that appears and explain, in terms of what is being neutralised, why **Equivalence vol.** shows a dash.
 - Drop both concentrations to **0.020 M** with a weak acid at pK_a 4.75. The equivalence volume does not move, but the vertical jump shrinks. Which indicators from Part 4 still land inside it?
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